

CVaR-Based Portfolio Optimisation for a Malta-Accessible UCITS ETF Basket

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Abstract—This paper offers some additional guidelines for MCAST IICT 2nd year B.Sc. students on research paper writing. The abstract is the first paragraph that any researcher reads and thus you need to capture the essence of your research. Dedicate 1 sentence for the theme (subject matter), another for your project aim, one for the proposed solution, then for the general outcome of this project (positive/negative result/outcome), another for the technology used (Windows/Linux, Android/iOS, C#/Java, Unity/Unreal), finally for any technique used (Neural Networks, Pearl Noise, Market Basket Analysis, K-Means, HMM, Augmented Reality). Following is an example: *In this research we are tackling the automatic annotation of cast members and use of key props within movies. Movies tend to gather a huge fan gathering who dedicate a lot of time and resources for documenting the content, thus tools to automate processes such as the timeline when cast members appear, presents of props such as weapons and other forms of annotations are desired. In this research we propose the use of image processing techniques, namely Principal Component Analysis (PCA) and Convolutional Neural Networks (CNN) for the automatic annotation of actors' timeline and presence of weapons in the popular Tv series Game of Thrones (GoT). Our proposed solution managed to detect the targets in 85% of the cases and we identify situations where this is challenging and recommend future research directions.*

Index Terms—MCAST, IICT, L^AT_EX, Project, Paper

I. INTRODUCTION

Recent market crises, including the COVID-19 shock, have renewed criticism of classical portfolio theory. Although the mean-variance (MV) framework remains foundational, variance treats upside and downside deviations symmetrically and therefore does not isolate severe losses in the way downside-focused investors typically require [1], [2]. For this reason, portfolio research has increasingly adopted Conditional Value-at-Risk (CVaR), which estimates average losses beyond the VaR threshold and can be incorporated into tractable optimisation formulations [2], [3]. Exchange-traded funds (ETFs) provide a practical setting for such comparison because prior studies have evaluated CVaR-based, MV, and equal-weight approaches using diversified ETF universes and out-of-sample backtesting [3].

Despite this relevance, much of the recent CVaR literature is embedded in advanced settings such as evolutionary algorithms, copula-based dependence modelling, bilevel multi-market optimisation, deep reinforcement learning, and distributionally robust optimisation [2]–[6]. These studies are

valuable, but they make it harder to judge whether a simpler and more transparent CVaR-minimisation framework, implemented under standard long-only constraints, already provides meaningful downside-risk benefits. In addition, while Malta has an established collective investment scheme environment and the Maltese Assets Fund illustrates the local relevance of diversified fund products [7], [8], there is limited evidence on CVaR-based optimisation in a Malta-accessible UCITS ETF context. The present study is therefore positioned at the intersection of quantitative portfolio theory and applied investment management, addressing a contextual and empirical gap by testing whether a Malta-accessible UCITS ETF basket can serve as a credible benchmark setting for comparing CVaR, Equal-Weight, and Minimum-Variance strategies.

The purpose of this quantitative backtesting study is to evaluate whether CVaR-minimisation provides stronger downside-risk control than Equal-Weight (1/N) and Minimum-Variance portfolio construction in a Malta-accessible UCITS ETF basket over 2016–2025. The independent variable is the optimisation strategy, operationalised through three portfolio rules: Equal-Weight, Minimum-Variance, and CVaR-minimisation. The dependent variables are the observed portfolio outcomes, namely CAGR, volatility, Sharpe ratio, Sortino ratio, VaR, CVaR, and maximum drawdown. To preserve comparability, the ETF universe, estimation window length, rebalancing frequency, and long-only fully invested constraints are treated as control variables. The study aims to implement the three strategies under identical conditions using rolling out-of-sample backtesting and to compare their performance on both return-based and downside-risk criteria.

Accordingly, the main hypothesis states that a CVaR-minimised portfolio will achieve lower CVaR and lower maximum drawdown than Equal-Weight and Minimum-Variance portfolios without material deterioration in CAGR or Sharpe ratio. The null hypothesis states that no meaningful difference will exist among the three strategies in CVaR, maximum drawdown, or risk-adjusted performance. As shown in Fig. 1, the present study adopts a positivist philosophy and a deductive approach. It employs a controlled experimental backtesting strategy using secondary historical ETF data, a mono-method quantitative design, and a longitudinal time horizon. The techniques and procedures consist of secondary ETF data

collection, rolling out-of-sample backtesting, and quantitative performance and downside-risk analysis. Specifically, portfolio outcomes are treated as objectively measurable through predefined financial metrics. A deductive approach is adopted because the study begins from established theory on the limitations of variance as a risk proxy and tests whether a downside-risk measure produces superior empirical outcomes [1], [2]. Since performance is evaluated across repeated windows over 2016–2025 rather than at a single point in time, the time horizon is longitudinal.

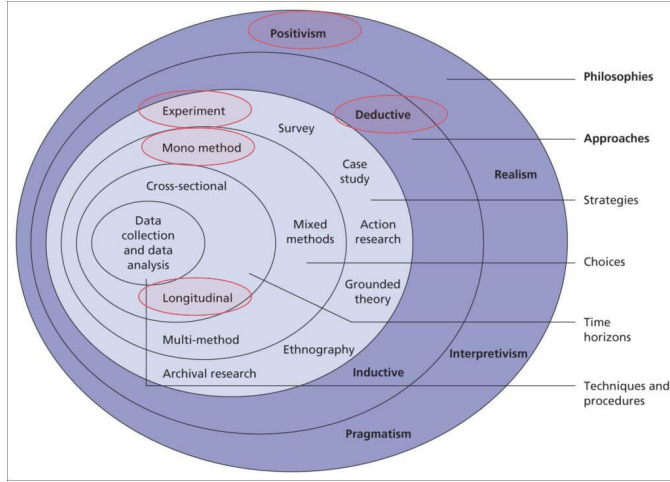


Fig. 1. Research onion for the present study, adapted from Fig. 4.1 in Saunders, Lewis, and Thornhill [9].

II. LITERATURE REVIEW

Give a **brief** overview of the theme, history and facts. You should use the present tense since what you are documenting is current knowledge. Do not dedicate a lot to historical facts about the subject matter, but limit to a paragraph. Make sure you quantify by quoting statistics where relevant. Following is an example for a research in “Automatic cyber-bullying detection”: *With the popularity of social media, predominantly amongst the adolescent and young adult generations, there appears to be a correlation between the increase in suicide rates and social networking adoption, with suicide attempts increasing by 10% amongst adolescents and by 5.6% amongst young adults for the period 2013-2017.*

Once you have documented some background on the theme you need to focus on three key aspects: **Data set**; **Algorithms**; **Evaluation**. So you need to research existing data sets and document them. Say we want to explore accent recognition in audio/video clips, you would need to identify existing data sets and document first by enumerating then by comparing in a table layout such as in Table I. A useful tool for creation of tables is Truben’s Table Tool¹. When enlisting the data sets make sure you provide the proper citation. When comparing you need to identify the key features/factors/variable that are important for your subject matter.

¹<http://truben.no/table/>

- 1) The Accents of the British Isles (ABI-1) Speech Corpus, /cited2004accents
- 2) TIMIT Acoustic-Phonetic Continuous Speech Corpus, /citegarofalo1993darpa
- 3) Common Voice Data Set, /cite
- 4) WSJCAM0 Corpus, /cite
- 5) The Speech Accent Archive, /citeweinberger2011speech
- 6) The Arctic CMU Audio Database, citekominek2004cmu
- 7) VoxCeleb2, citechung2018voxceleb2

TABLE I
DATA-SET COMPARISON. ACCENT: NATIVE (N); NON-NATIVE (NN)

Data set	Speakers	Utterances	Accents	Citations	Type
1	280	855	14	28	N
2	630	6,300	8	1,918	N
3	50,590	644,120	16	N/A	N & NN
4	140	4,400	4	346	N
5	2,140	2,140	>100	132	N & NN
6	18	1,150	4	458	N & NN
7	6,000	1,128,246	6	168	N & NN

Next you need to provide an overview of what researchers have proposed and their key results. A survey or review paper would be ideal since these kind of papers would cover multiple techniques and provide good comparisons. Returning to our example of “Automatic cyber-bullying detection” consider the technique comparison documented in Table II. Note the structure used in this table, first you have the citation of the research, followed by the data set used, then the metric quoted for comparison, finally the relevant result.

TABLE II
STATE OF THE ART RESULTS

Study	Data set	Metrics	Results
/citeprathyusha2017cyberbully	Crime Investigation Forum	F1 Score	CDMS: 40.75% CDCSGF: 39.75%
/citevan2018automatic	Corpus for English and Dutch	F1 Score	English: 64% Dutch: 61%
/citeal2016cybercrime	Twitter	F1 Score	93.6%
/citezhong2016content	Instagram	F1 Score	SVM Classifier: 95.00% Image Classifier: 68.08%
/citereynolds2011using	Formspring	Accuracy	78.5%
/citedinakar2011modeling	Youtube	Accuracy	66.70%

The final recommended part in a good literature review is a review in the evaluation metrics used. It is not possible to cover all possible evaluation metrics but in most cases there is a classification problem, which would require something called a confusion matrix. Consider the problem of recognising and distinguishing an image of a cat from that of a dog. You will have a data set split into training and testing. Both will have the ground truth, what the image is actually showing. This ground truth, in the case of the training data set will be used for the creation of the Artificial Intelligence (AI) model, whilst the ground truth in the testing data set will be used to compare the predicted classification. The predictions are plotted in a confusion matrix as shown in Table III. The same table can be extended to have more than 2 categories such as introducing the mouse category. Then you need to cycle the focus for each category as shown in Table IV and Table V, then extract the key values namely:

TABLE III
SAMPLE CAT VS DOG CONFUSION MATRIX

		Predicted	
		Cat	Dog
Actual	Cat	33	2
	Dog	5	30

TABLE IV
CAT CATEGORY EVALUATION METRICS

		Predicted	
		Cat	Not-Cat
Actual	Cat	TP	FN
	Not-Cat	FP	TN

- 1) **True Positives (TP)** - This refers to the correctly predicted positive values i.e. the value of both the actual class and predicted class is yes.
- 2) **True Negatives (TN)** - These are the correctly predicted negative values i.e. the value of both actual class and predicted class is no.
- 3) **False Positives (FP)** - It indicates values where the actual class is no but the predicted class is yes.
- 4) **False Negatives (FN)** - It indicates values where the actual class is yes but the predicted class is no.

The confusion matrices are used to provide evaluation metrics such as Accuracy, Precision, Recall and F1-Score. Depending on the subject matter being researched the proper metrics need to be quoted, thus the need to research this in academic literature. For a quick but reliable reference you can consider Wikipedia's dedicated page².

- 1) **Accuracy (ACC)** - This shows the measure of effectiveness of the machine learning model.

$$Accuracy(ACC) = \frac{TP + TN}{P + N} \quad (1)$$

- 2) **Precision (PPV)** - Precision is the ratio of correctly predicted positive values to the total predicted positive values. This metric highlights the correct positive predictions out of all the positive predictions. High precision indicates low false positive rate.

$$Precision(PPV) = \frac{TP}{TP + FP} \quad (2)$$

- 3) **Recall (TPR)** - The recall is the ratio of correctly predicted positive values to the actual positive values. Recall highlights the sensitivity of the algorithm i.e. out

²https://en.wikipedia.org/wiki/Confusion_matrix

TABLE V
DOG CATEGORY EVALUATION METRICS

		Predicted	
		Not-Dog	Dog
Actual	Not-Dog	TN	FP
	Dog	FN	TP

of all the actual positives how many were caught by the program.

$$Recall(TPR) = \frac{TP}{P} \quad (3)$$

- 4) **F1 Score** - F1 Score is the weighted average of Precision and Recall. Therefore, this score takes both false positives and false negatives into account. Intuitively it is not as easy to understand as accuracy, but F1 is usually more useful than accuracy, especially if you have an uneven class distribution.

$$F1Score = \frac{2TP}{2TP + FP + FN} \quad (4)$$

III. RESEARCH METHODOLOGY

This section is one of the most important parts of your research and it is the one that will receive most criticism. The reason is that here you document how you conducted your research. Since you are reading a B.Sc. degree you are expected to follow a scientific method, thus in this section you are expected to document: **the problem; hypothesis; aim & objectives; research questions; and research pipeline.** Let's go through each.

First start by providing a brief overview of the problem, similar to what you already mentioned in the previous sections. Then provide a hypothesis, this is the assumption you are making, which via your research you are attempting to prove or disprove. An example hypothesis in traffic management would be: *By using image processing on public camera video feeds we believe that it is possible to estimate the traffic congestion at a higher precision than public traffic data feeds.*

After explaining the problem and hypothesis you need to specify why you are doing this research and what you think the output will be, so the aim & objectives. Let us consider the "Marine Vessel Tracking" concept: *With this research we aim to be able to automatically detect any anomalies within Marine traffic, such as drug smuggling, human trafficking and/or collisions.* Our objectives are:

- 1) Identify existing data sets and manners of connecting them or even enriching them.
- 2) Review existing algorithms and propose a suitable model
- 3) Evaluate the effectiveness of the proposed model with real data and existing research.

So lastly you have to present a number of research questions that you attempt to answer, which ideally revolve around the **data set, algorithm and evaluation** which you documented in your literature review:

- 1) Does a data set need to be created for such a research?
- 2) Which current technology is best suited to address the problem?
- 3) How does the proposed solution compare with existing solutions?

Finally, you need a plan on how you intend to answer the research questions which will help you prove/disprove the hypothesis. Different research areas would require specific pipelines, yet a generic one is shown in Fig 2. After presenting

an illustration of your pipeline it is recommended to document what has been done at every stage. Remember you should use past tense in this section since you are document what you have already done, since this paper will be published at the end of your research.

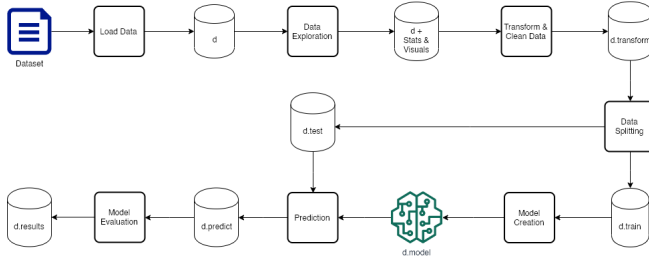


Fig. 2. Research Pipeline

IV. FINDINGS & DISCUSSION OF RESULTS

This section is a delicate section and you are encouraged to be as honest and open as possible. The aim is not to show that you got a perfect solution to a long existing problem. Trying to state that in a few months you developed a perfect solution with 100% is not convincing and raises doubts. It is recommended that you document your findings in every step of your research pipeline, highlighting your observations and decisions taken. Present your results and very importantly compare with existing research which you documented in Section II.

The focus of the Project module is for you to delve into an area that exposes you to new technologies and offers you an opportunity to be critical of your work. So you are expected to document where your solution/research worked and where it did not. Reflect and document reasons why the solution/research did not perform as expected and propose ways of addressing this. From these observations you will produce new research questions in the next section. Consider the research in “Brand usage detection within audio streams”, where certain key terms were searched within videos, the results of which are documented in Table VI. You will notice that the terms “Peppa” and “Sushi” were the least recognised terms even by the best transcribers. Upon investigation we determined that “Peppa” was not recognised cause of voice morphing to create childish voices in the cartoon video, whilst “Sushi” was pronounced by a Japanese person speaking English. So the research student decided to focus his dissertation research on how to create a system that is able to recognise heavy accents to automate the configuration of a transcriber, in this case to cater for English spoken by a Japanese person, which accent is very different from an Indian accent, British accent or Italian accent, just to name a few.

V. CONCLUSION

So the conclusion is most probably the second section that a reader would use to consider reading in full your research. Thus it is important to highlight the essence of your research. The recommended approach is to answer your research

TABLE VI
RECALL RESULTS

#	Term	Google Cloud	Google Speech	Sphinx CMU
1	Peppa	27%	33%	0%
2	Peppa	33%	22%	0%
3	Apple	96%	92%	79%
4	Galaxy	100%	100%	100%
5	Galaxy	95%	95%	80%
6	Sushi	75%	35%	0%
	Average	71%	62%	43%

methodology. Start by answering your research questions, then stating to what degree did this research achieve its aim and objectives, highlighting potential causes for not being able to do so at a desired level, such as time, or other circumstances. Consider the following: *A student was due to research the use of MCAST computers during out-of-office hours to offer a private cloud computing service for research, similar to Google Cloud but free for MCAST students. Due to the lockdown by COVID-19 pandemic we could not continue on the original planned research objective and had to adapt.*

The final and most important part of the conclusion are your recommendations for future research, not necessarily for yourself (referring to what you plan to do in your dissertation or beyond), but also to other future researchers who might consider doing similar work to yours. The recommendations you provide here will set such prospective researchers on a better track/direction thanks to your experience.

APPENDIX A SUPPORTING MATERIAL

You can add screen shots and statistics. Stick to essential information.

ACKNOWLEDGEMENT

You can dedicate this section for special assistance that you were given by a 3rd party. Not your mentor, relatives or questionnaire participants.

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